TidyTuesday Week 44

Pranay Gundam

Sunday 1st November, 2026

Contents

1 Weekly Summary **2**

2 Date: 2026-10-26 **4**
- 2.1 Raw Regression . . . 4
- 2.2 Detrended Regression . . . 6

3 Date: 2026-10-27 **8**
- 3.1 Raw Regression . . . 8
- 3.2 Detrended Regression . . . 10

4 Date: 2026-10-28 **12**
- 4.1 Raw Regression . . . 12
- 4.2 Detrended Regression . . . 14

5 Date: 2026-10-29 **16**
- 5.1 Raw Regression . . . 16
- 5.2 Detrended Regression . . . 18

6 Date: 2026-10-30 **20**
- 6.1 Raw Regression . . . 20
- 6.2 Detrended Regression . . . 22

7 Date: 2026-10-31 **24**
- 7.1 Raw Regression . . . 24
- 7.2 Detrended Regression . . . 26

8 Date: 2026-11-01 **28**
- 8.1 Raw Regression . . . 28
- 8.2 Detrended Regression . . . 30

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Another glorious week of shoving two complete strangers into a regression and asking them to explain each other's life choices. As always, the official disclaimer up front: these pairings are chosen by the cold, uncaring hand of a random number generator, nobody here is claiming anything causes anything, and if you walk away believing Danish solar stocks control Polish recessions, that is a you problem, not a statistics problem. Spurious is the house style. Let us proceed.

Monday kicked things off with the OMX Copenhagen Alternative Energy Price Index versus a discontinued Polish recession indicator, which is already the kind of blind date that ends with both parties checking their phones. The raw regression coughed up a technically-significant p-value attached to an R-squared of roughly 0.06, which is the quantitative equivalent of "we have literally nothing in common but sure, we kissed." Amazingly, the detrended version actually held up and even nudged the R-squared up to 0.07, so this one survived the trend-stripping gauntlet. I would love to tell you this means something. It means two series explain seven percent of each other. Hold your applause. Tuesday was mercifully honest: car-part export prices versus Chilean purchases of U.S. agency bonds produced a p-value of 0.43 raw and a majestic 0.90 detrended, which is the number you get when two series refuse to acknowledge each other's existence. Beautiful. No notes.

Then came the midweek festival of trend mirages, our bread and butter. Wednesday paired consumer motor-vehicle leases with foreign-held mortgage-backed securities and posted a show-stopping raw R-squared of 0.81 – gasp! – only for the detrended version to deflate to 0.44. Still significant, which is suspicious enough to be mildly interesting, but that thirty-seven-point faceplant is pure "they both went up over time" energy. Thursday did the same trick harder: Covered Employment versus a Midwest house price index strutted in at 0.79 raw, then collapsed to 0.17 once we took away its shared-drift crutch, which is the statistical arc of a soufflé. Friday was the purest mirage of the week – child poverty in Leon County, Florida versus high-tier San Francisco home prices, a gaudy 0.42 raw that detrended all the way down to 0.006 with a p-value of 0.71. That is not a relationship collapsing; that is a relationship that was never there filing for divorce.

Saturday gave us the week's most entertaining betrayal: Mid-Atlantic private employment versus a management-compensation cost index, a towering 0.77 raw R-squared that evaporated to 0.009 detrended, p-value 0.44. Two series that both trend upward because the economy tends to do that, caught red-handed. And Sunday closed us out with retail sales percent-change versus

San Angelo, Texas home prices, which never even bothered to fake it – 0.005 raw, 0.004 detrended, both profoundly insignificant. A fittingly humble finish. So the week's scoreboard: a parade of raw "hits" that crumbled the instant we asked them to prove themselves, and Monday's weird little copper-colored survivor that technically held up while explaining almost nothing. Treat all of it as the random noise it proudly is, and I will see you next week for more of the same.

2 Date: 2026-10-26

Series ID: NASDAQCX601020PI
This series is titled OMX Copenhagen Alternative Energy Price Index and has a frequency of Daily. The units are Index and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2008-12-30 and the observation end date is 2026-10-23.The popularity of this series is 1.

Series ID: POLRECDM
This series is titled OECD based Recession Indicators for Poland from the Peak through the Trough (DISCONTINUED) and has a frequency of Daily, 7-Day. The units are +1 or 0 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1995-02-01 and the observation end date is 2022-08-31.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_POLRECDM	**R-squared:**	0.057
Model:	OLS	**Adj. R-squared:**	0.057
Method:	Least Squares	**F-statistic:**	206.4
Date:	Mon, 26 Oct 2026	**Prob (F-statistic):**	1.74e-45
Time:	17:43:09	**Log-Likelihood:**	-2167.1
No. Observations:	3419	**AIC:**	4338.
Df Residuals:	3417	**BIC:**	4351.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.4795	0.013	36.620	0.000	0.454	0.505
value_fred_NASDAQCX601020PI	-4.034e-05	2.81e-06	-14.366	0.000	-4.58e-05	-3.48e-05

Omnibus:	47995.954	**Durbin-Watson:**	0.009
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	514.195
Skew:	0.676	**Prob(JB):**	2.21e-112
Kurtosis:	1.665	**Cond. No.**	7.83e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.83e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

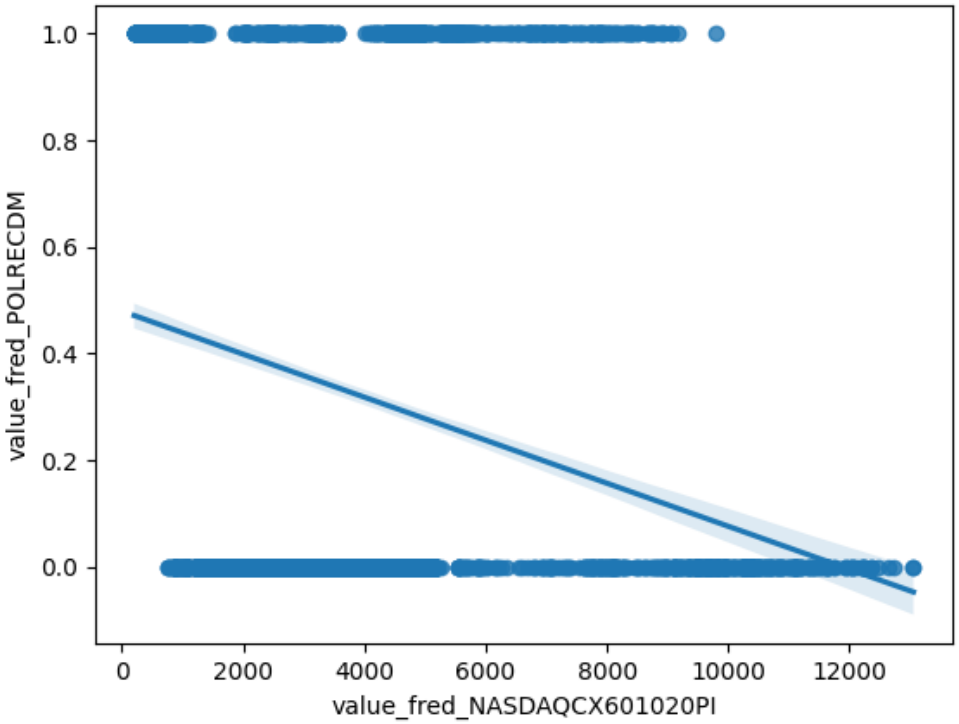


Figure 1: Raw Regression Plot for 2026-10-26

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_POLRECDM_detrended	**R-squared:**	0.071
Model:	OLS	**Adj. R-squared:**	0.071
Method:	Least Squares	**F-statistic:**	262.7
Date:	Mon, 26 Oct 2026	**Prob (F-statistic):**	5.73e-57
Time:	17:43:09	**Log-Likelihood:**	-2123.7
No. Observations:	3419	**AIC:**	4251.
Df Residuals:	3417	**BIC:**	4264.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.884e-14	0.008	-3.74e-12	1.000	-0.015	0.015
value_fred_NASDAQCX601020PI_detrended	-7.539e-05	4.65e-06	-16.207	0.000	-8.45e-05	-6.63e-0

Omnibus:	113787.732	**Durbin-Watson:**	0.009
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	482.553
Skew:	0.631	**Prob(JB):**	1.64e-105
Kurtosis:	1.660	**Cond. No.**	1.66e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.66e+03. This might indicate that there are strong multicollinearity or other numerical problems.

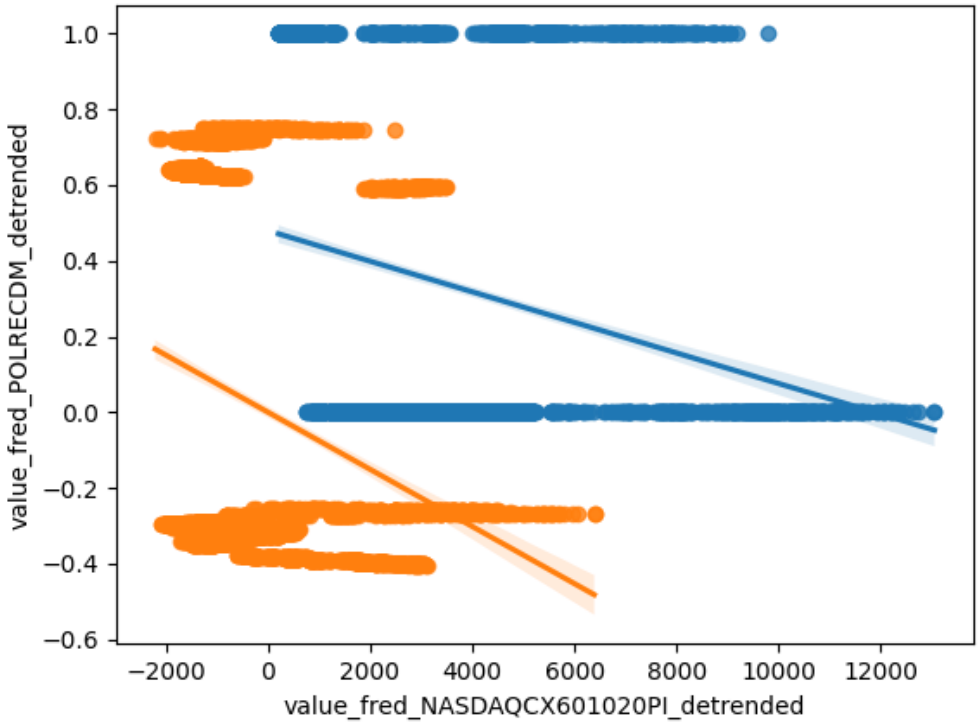


Figure 2: Detrended Regression Plot for 2026-10-26

3 Date: 2026-10-27

Series ID: IY33632

This series is titled Export Price Index (NAICS): Motor Vehicle Electrical and Electronic Equipment Manufacturing and has a frequency of Monthly. The units are Index Dec 2008=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2008-12-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

Series ID: FORLTAGCYNET30406

This series is titled Foreign Net Transactions of U.S. Long-Term Agency Bonds: Chile and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1985-01-01 and the observation end date is 2026-08-01.The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FORLTAGCYNET30406	**R-squared:**	0.003
Model:	OLS	**Adj. R-squared:**	-0.002
Method:	Least Squares	**F-statistic:**	0.6151
Date:	Tue, 27 Oct 2026	**Prob (F-statistic):**	0.434
Time:	20:06:38	**Log-Likelihood:**	-1228.3
No. Observations:	212	**AIC:**	2461.
Df Residuals:	210	**BIC:**	2467.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	76.4398	92.089	0.830	0.407	-105.098	257.978
value_fred_IY33632	-0.7484	0.954	-0.784	0.434	-2.630	1.133

Omnibus:	296.720	**Durbin-Watson:**	2.046
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	32875.754
Skew:	6.102	**Prob(JB):**	0.00
Kurtosis:	62.773	**Cond. No.**	1.62e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.62e+03. This might indicate that there are strong multicollinearity or other numerical problems.

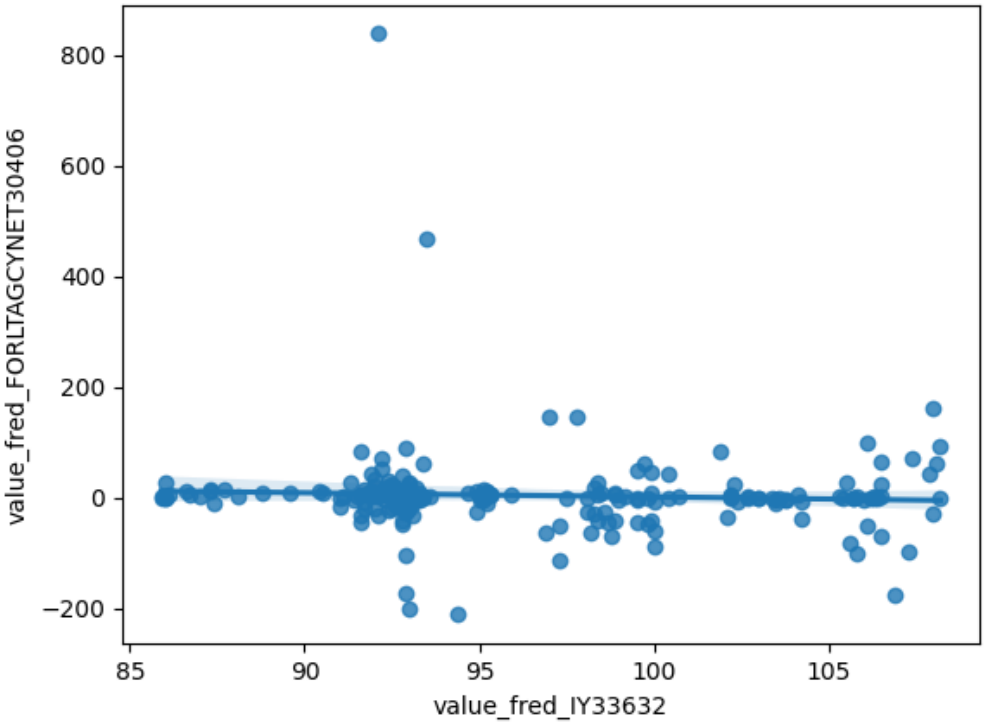


Figure 3: Raw Regression Plot for 2026-10-27

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FORLTAGCYNET30406_detrended	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	-0.005
Method:	Least Squares	**F-statistic:**	0.01566
Date:	Tue, 27 Oct 2026	**Prob (F-statistic):**	0.901
Time:	20:06:39	**Log-Likelihood:**	-1228.1
No. Observations:	212	**AIC:**	2460.
Df Residuals:	210	**BIC:**	2467.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.565e-12	5.476	2.86e-13	1.000	-10.794	10.794
value_fred_IY33632_detrended	-0.1687	1.348	-0.125	0.901	-2.825	2.488

Omnibus:	295.354	**Durbin-Watson:**	2.049
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	32265.413
Skew:	6.055	**Prob(JB):**	0.00
Kurtosis:	62.212	**Cond. No.**	4.06

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

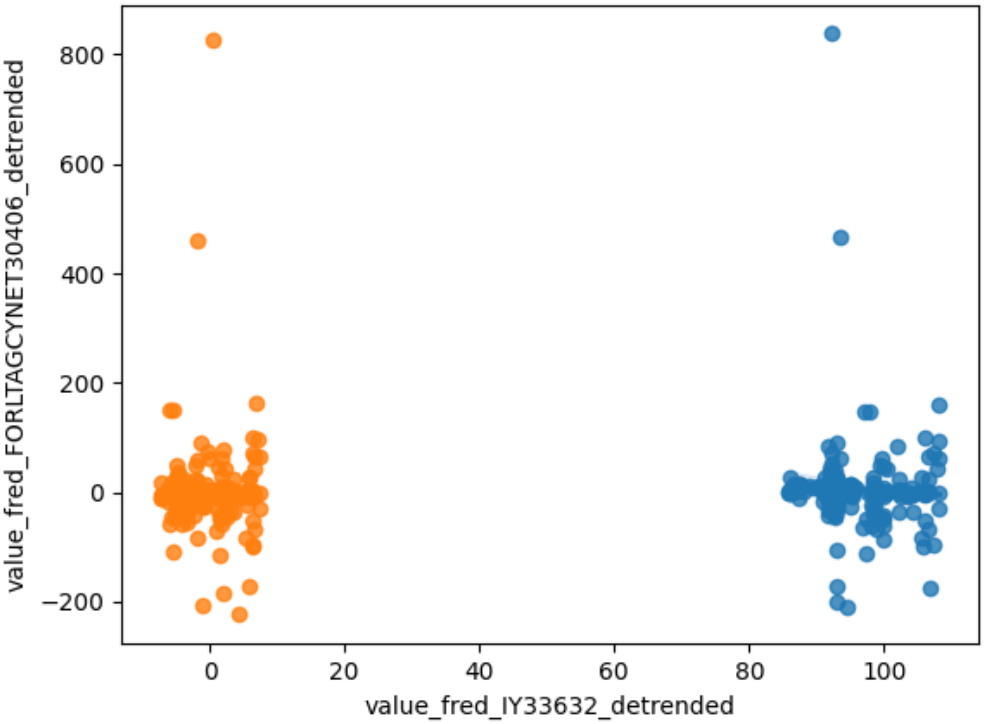


Figure 4: Detrended Regression Plot for 2026-10-27

4 Date: 2026-10-28

Series ID: DTCORVHFNM
This series is titled Consumer Motor Vehicle Leases Owned by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1980-06-01 and the observation end date is 2026-07-01.The popularity of this series is 13.

Series ID: OMBFRIM027SBOG
This series is titled Other Securities: Mortgage-Backed Securities, Foreign-Related Institutions and has a frequency of Monthly. The units are Billions of U.S. Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 2009-07-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OMBFRIM027SBOG	**R-squared:**	0.811
Model:	OLS	**Adj. R-squared:**	0.810
Method:	Least Squares	**F-statistic:**	872.2
Date:	Wed, 28 Oct 2026	**Prob (F-statistic):**	2.03e-75
Time:	17:29:33	**Log-Likelihood:**	-373.28
No. Observations:	205	**AIC:**	750.6
Df Residuals:	203	**BIC:**	757.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	17.1997	0.459	37.497	0.000	16.295	18.104
value_fred_DTCORVHFNM	-7.456e-05	2.52e-06	-29.533	0.000	-7.95e-05	-6.96e-05

Omnibus:	20.349	**Durbin-Watson:**	0.152
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	23.791
Skew:	0.816	**Prob(JB):**	6.82e-06
Kurtosis:	2.650	**Cond. No.**	7.94e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.94e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

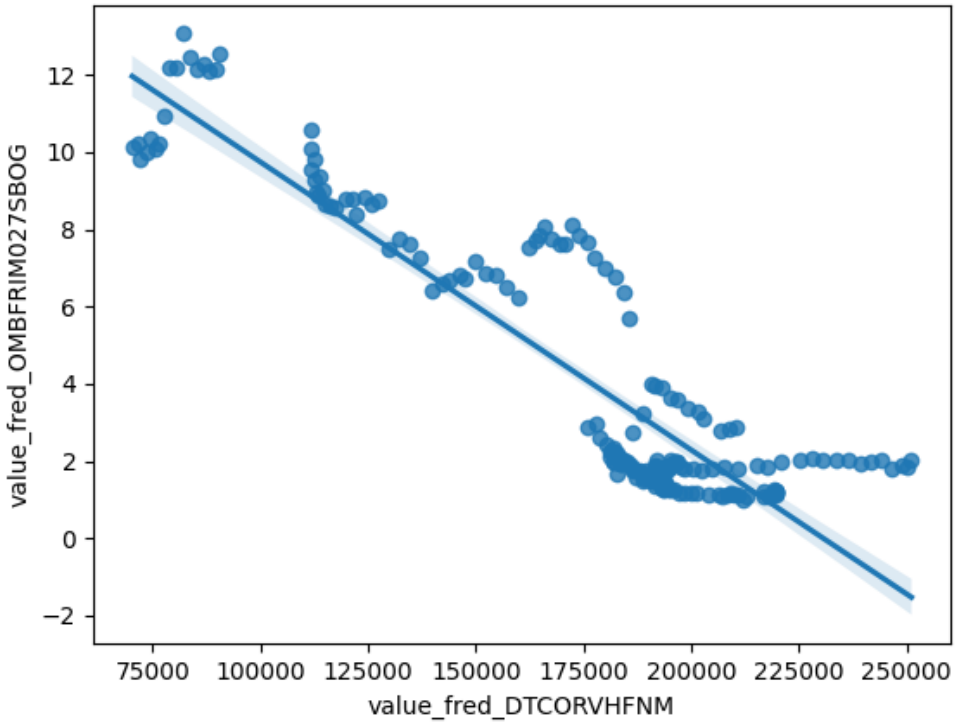


Figure 5: Raw Regression Plot for 2026-10-28

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OMBFRIM027SBOG_detrended	**R-squared:**	0.445
Model:	OLS	**Adj. R-squared:**	0.442
Method:	Least Squares	**F-statistic:**	162.8
Date:	Wed, 28 Oct 2026	**Prob (F-statistic):**	9.21e-28
Time:	17:29:33	**Log-Likelihood:**	-348.91
No. Observations:	205	**AIC:**	701.8
Df Residuals:	203	**BIC:**	708.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-1.902e-12	0.093	-2.04e-11	1.000	-0.184	0.184
value_fred_DTCORVHFNM_detrended	-5.049e-05	3.96e-06	-12.759	0.000	-5.83e-05	-4.27e-05

Omnibus:	22.829	**Durbin-Watson:**	0.125
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	8.351
Skew:	0.217	**Prob(JB):**	0.0154
Kurtosis:	2.112	**Cond. No.**	2.35e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.35e+04. This might indicate that there are strong multicollinearity or other numerical problems.

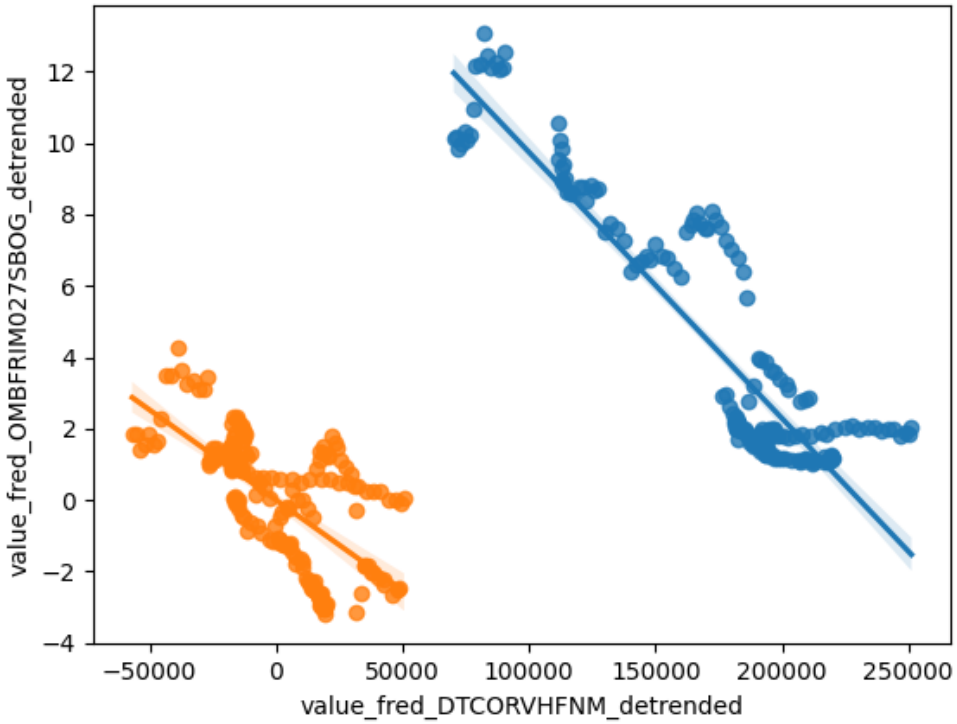


Figure 6: Detrended Regression Plot for 2026-10-28

5 Date: 2026-10-29

Series ID: COVEMP
This series is titled Covered Employment and has a frequency of Weekly, Ending Saturday. The units are Number and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1971-01-02 and the observation end date is 2026-10-10.The popularity of this series is 16.

Series ID: PONHPI00106M226N
This series is titled Purchase Only House Price Index for the East North Central Census Division and has a frequency of Monthly. The units are Index Jan 1991=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1991-01-01 and the observation end date is 2026-08-01.The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PONHPI00106M226N	**R-squared:**	0.787
Model:	OLS	**Adj. R-squared:**	0.784
Method:	Least Squares	**F-statistic:**	218.3
Date:	Thu, 29 Oct 2026	**Prob (F-statistic):**	1.72e-21
Time:	17:40:13	**Log-Likelihood:**	-300.50
No. Observations:	61	**AIC:**	605.0
Df Residuals:	59	**BIC:**	609.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-444.7300	43.423	-10.242	0.000	-531.619	-357.841
value_fred_COVEMP	4.924e-06	3.33e-07	14.776	0.000	4.26e-06	5.59e-06

Omnibus:	6.707	**Durbin-Watson:**	0.108
Prob(Omnibus):	0.035	**Jarque-Bera (JB):**	4.815
Skew:	0.551	**Prob(JB):**	0.0900
Kurtosis:	2.174	**Cond. No.**	1.30e+09

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.3e+09. This might indicate that there are
strong multicollinearity or other numerical problems.

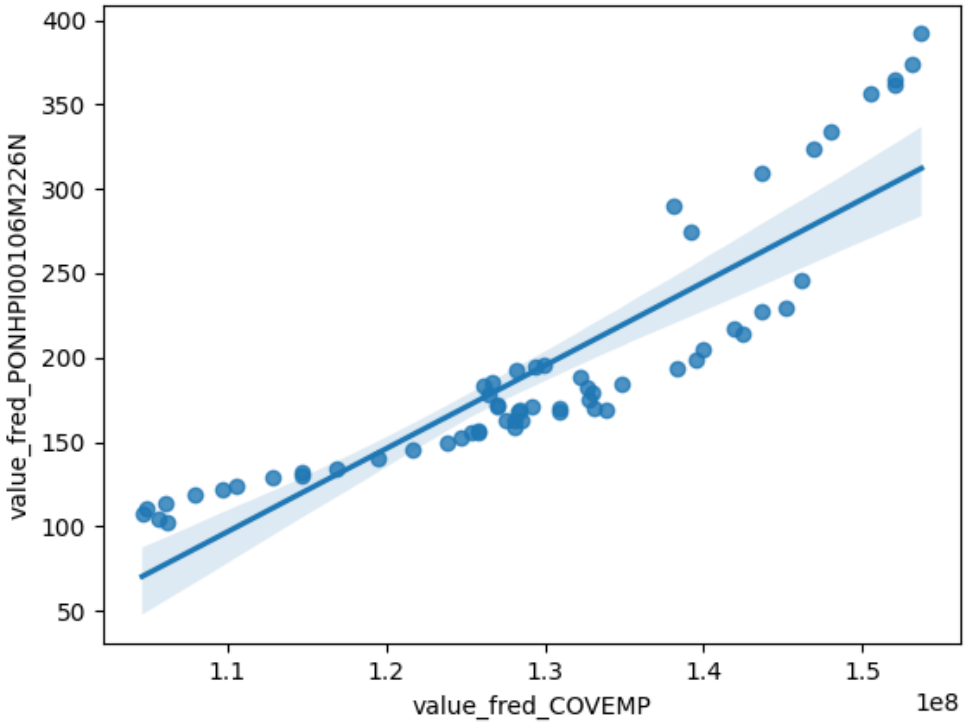


Figure 7: Raw Regression Plot for 2026-10-29

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PONHPI00106M226N_detrended	**R-squared:**	0.170
Model:	OLS	**Adj. R-squared:**	0.156
Method:	Least Squares	**F-statistic:**	12.07
Date:	Thu, 29 Oct 2026	**Prob (F-statistic):**	0.000965
Time:	17:40:13	**Log-Likelihood:**	-299.68
No. Observations:	61	**AIC:**	603.4
Df Residuals:	59	**BIC:**	607.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.801e-12	4.285	1.12e-12	1.000	-8.574	8.574
value_fred_COVEMP_detrended	3.658e-06	1.05e-06	3.475	0.001	1.55e-06	5.76e-06

Omnibus:	4.556	**Durbin-Watson:**	0.081
Prob(Omnibus):	0.103	**Jarque-Bera (JB):**	3.381
Skew:	0.437	**Prob(JB):**	0.184
Kurtosis:	2.248	**Cond. No.**	4.07e+06

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.07e+06. This might indicate that there are strong multicollinearity or other numerical problems.

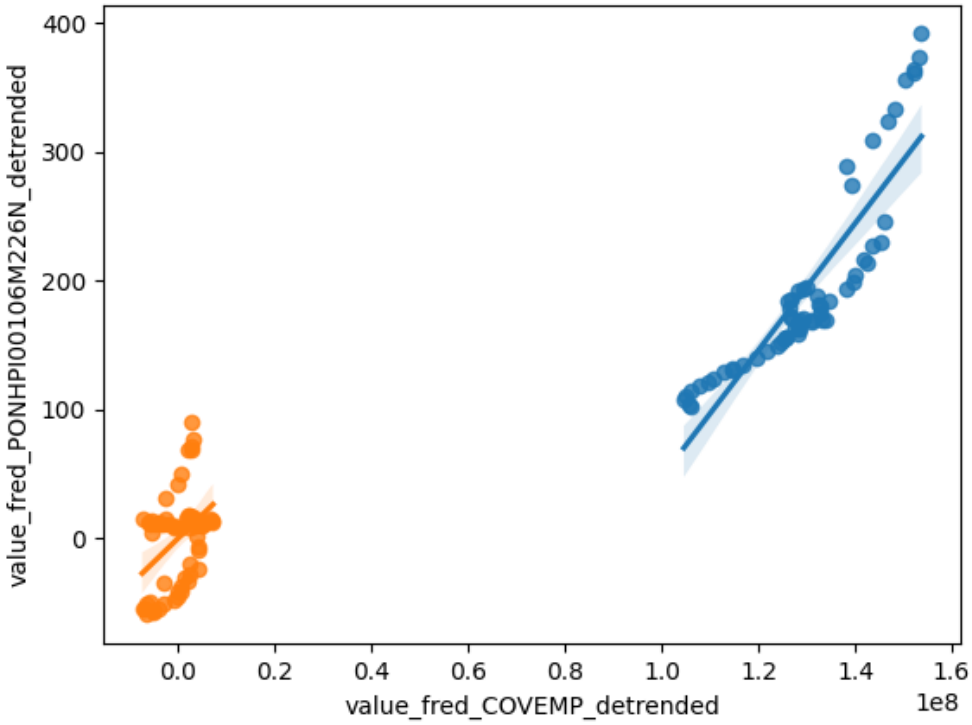


Figure 8: Detrended Regression Plot for 2026-10-29

6 Date: 2026-10-30

Series ID: PUA0T17FL12073A647NCEN
This series is titled Poverty Universe, Age 0-17 for Leon County, FL and has a frequency of Annual. The units are Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1998-01-01 and the observation end date is 2024-01-01.The popularity of this series is 1.

Series ID: SFXRHTSA
This series is titled Home Price Index (High Tier) for San Francisco, California and has a frequency of Monthly. The units are Index Jan 2000=100 and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1987-01-01 and the observation end date is 2026-08-01.The popularity of this series is 9.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SFXRHTSA	**R-squared:**	0.417
Model:	OLS	**Adj. R-squared:**	0.394
Method:	Least Squares	**F-statistic:**	17.87
Date:	Fri, 30 Oct 2026	**Prob (F-statistic):**	0.000276
Time:	17:18:43	**Log-Likelihood:**	-147.30
No. Observations:	27	**AIC:**	298.6
Df Residuals:	25	**BIC:**	301.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2231.4439	573.331	-3.892	0.001	-3412.242	-1050.646
value_fred_PUA0T17FL12073A647NCEN	0.0460	0.011	4.228	0.000	0.024	0.068

Omnibus:	2.121	**Durbin-Watson:**	0.831
Prob(Omnibus):	0.346	**Jarque-Bera (JB):**	0.876
Skew:	0.111	**Prob(JB):**	0.645
Kurtosis:	3.854	**Cond. No.**	2.67e+06

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.67e+06. This might indicate that there are
strong multicollinearity or other numerical problems.

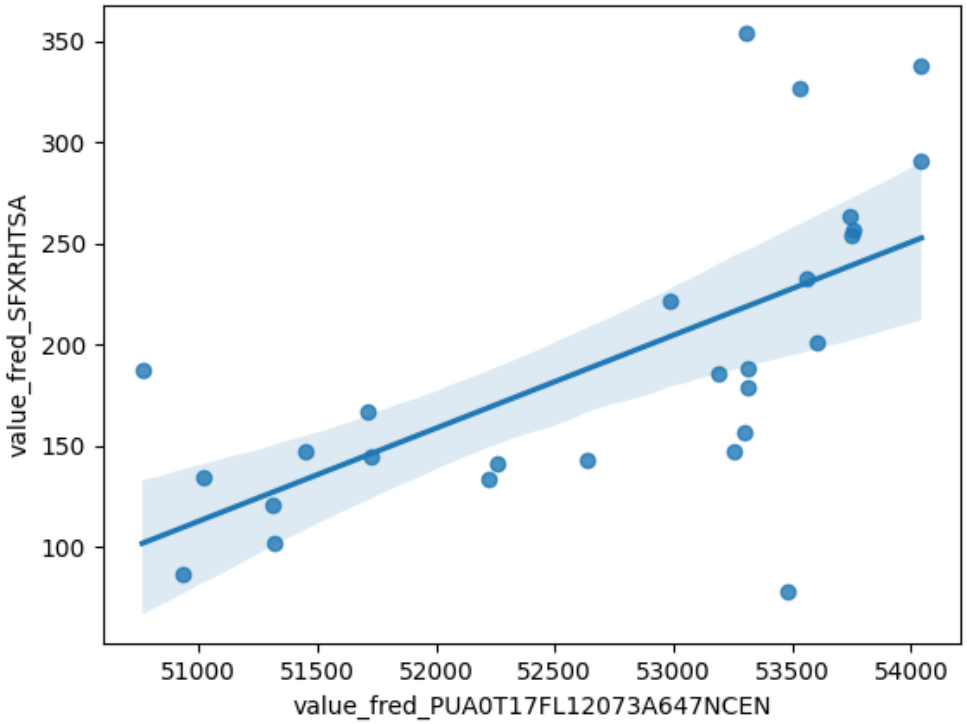


Figure 9: Raw Regression Plot for 2026-10-30

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SFXRHTSA_detrended	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	-0.034
Method:	Least Squares	**F-statistic:**	0.1395
Date:	Fri, 30 Oct 2026	**Prob (F-statistic):**	0.712
Time:	17:18:43	**Log-Likelihood:**	-129.54
No. Observations:	27	**AIC:**	263.1
Df Residuals:	25	**BIC:**	265.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.97**
const	-4.651e-12	5.867	-7.93e-13	1.000	-12.082	12.0
value_fred_PUA0T17FL12073A647NCEN_detrended	-0.0031	0.008	-0.374	0.712	-0.020	0.01

Omnibus:	0.333	**Durbin-Watson:**	0.403
Prob(Omnibus):	0.847	**Jarque-Bera (JB):**	0.398
Skew:	-0.231	**Prob(JB):**	0.819
Kurtosis:	2.624	**Cond. No.**	717.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

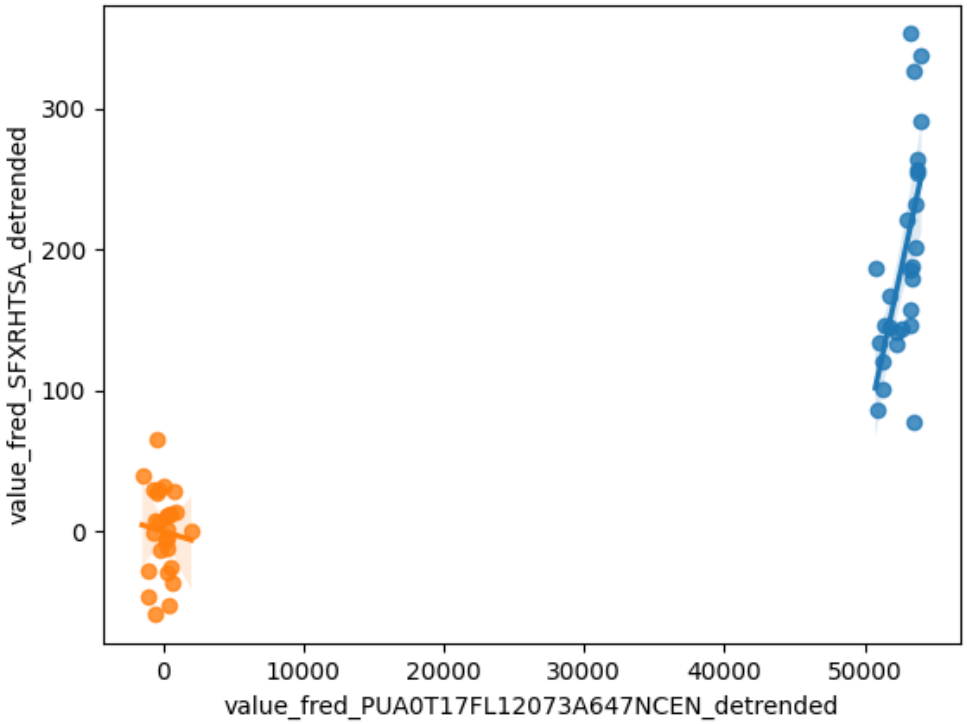


Figure 10: Detrended Regression Plot for 2026-10-30

7 Date: 2026-10-31

Series ID: ADPMCDMANERNSA
This series is titled Nonfarm Private Employment in the Middle Atlantic Census Division and has a frequency of Monthly. The units are Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2010-01-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

Series ID: CIU20100001000001
This series is titled Employment Cost Index: Total compensation for Private industry workers in Management, professional, and related and has a frequency of Quarterly. The units are Index Dec 2005=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2001-01-01 and the observation end date is 2026-07-01.The popularity of this series is 15.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CIU20100001000001	**R-squared:**	0.769
Model:	OLS	**Adj. R-squared:**	0.765
Method:	Least Squares	**F-statistic:**	216.4
Date:	Sat, 31 Oct 2026	**Prob (F-statistic):**	2.33e-22
Time:	16:24:36	**Log-Likelihood:**	-237.92
No. Observations:	67	**AIC:**	479.8
Df Residuals:	65	**BIC:**	484.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-153.4786	19.702	-7.790	0.000	-192.827	-114.130
value_fred_ADPMCDMANERNSA	1.793e-05	1.22e-06	14.709	0.000	1.55e-05	2.04e-05

Omnibus:	3.428	**Durbin-Watson:**	0.417
Prob(Omnibus):	0.180	**Jarque-Bera (JB):**	2.519
Skew:	0.319	**Prob(JB):**	0.284
Kurtosis:	2.297	**Cond. No.**	3.04e+08

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.04e+08. This might indicate that there are
strong multicollinearity or other numerical problems.

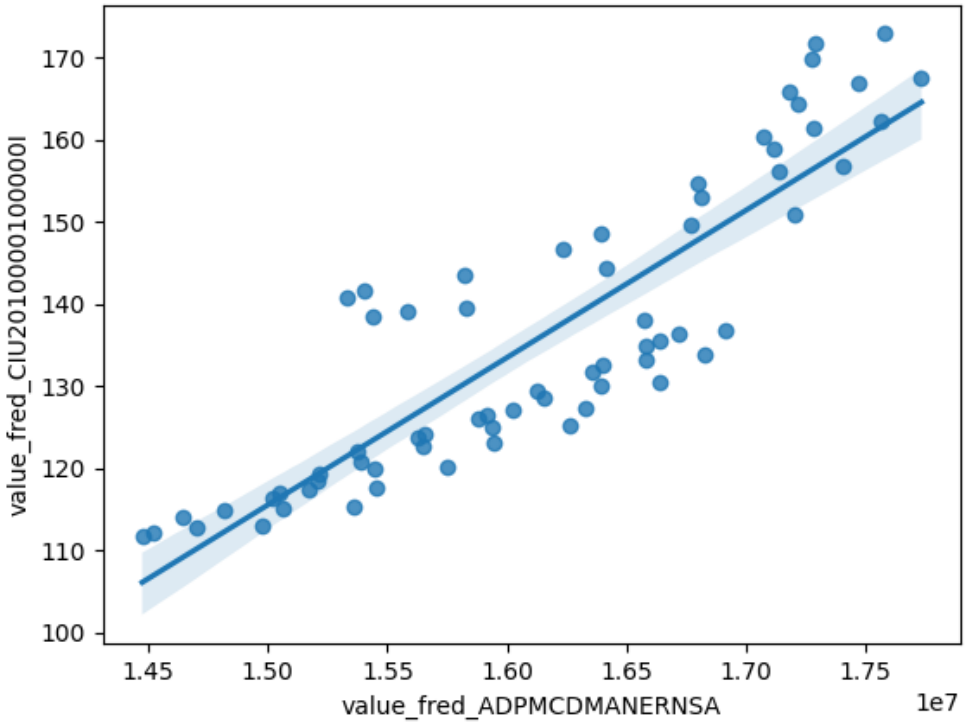


Figure 11: Raw Regression Plot for 2026-10-31

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CIU2010000100000I_detrended	**R-squared:**	0.009
Model:	OLS	**Adj. R-squared:**	-0.006
Method:	Least Squares	**F-statistic:**	0.5914
Date:	Sat, 31 Oct 2026	**Prob (F-statistic):**	0.445
Time:	16:24:37	**Log-Likelihood:**	-178.49
No. Observations:	67	**AIC:**	361.0
Df Residuals:	65	**BIC:**	365.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.132e-12	0.431	2.63e-12	1.000	-0.860	0.860
value_fred_ADPMCDMANERNSA_detrended	8.324e-07	1.08e-06	0.769	0.445	-1.33e-06	2.99e-06

Omnibus:	11.990	**Durbin-Watson:**	0.036
Prob(Omnibus):	0.002	**Jarque-Bera (JB):**	4.477
Skew:	0.343	**Prob(JB):**	0.107
Kurtosis:	1.936	**Cond. No.**	3.98e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.98e+05. This might indicate that there are strong multicollinearity or other numerical problems.

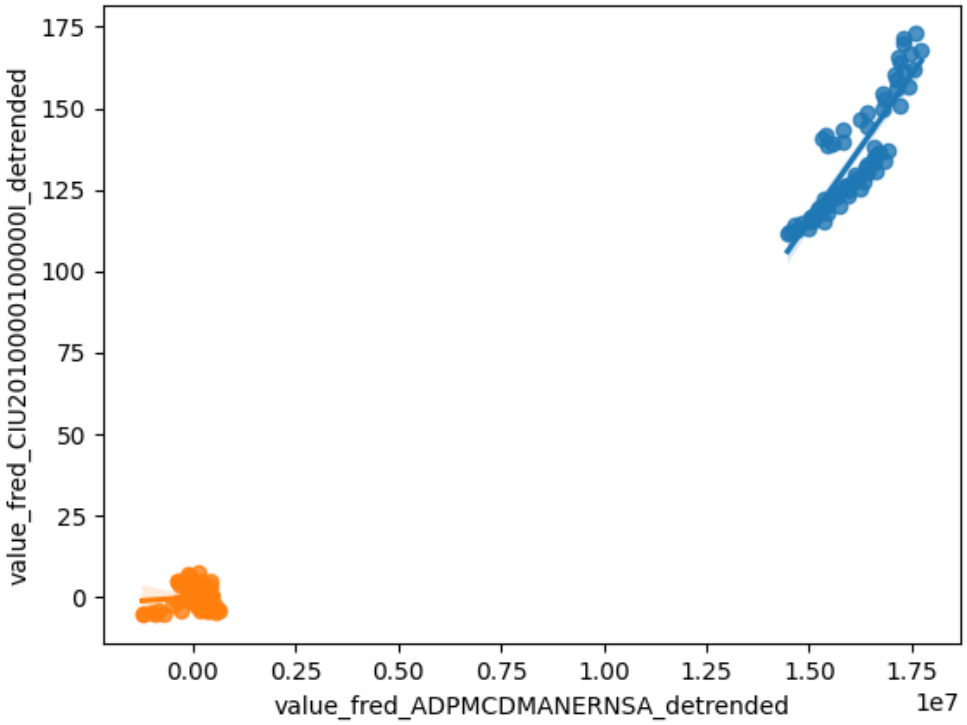


Figure 12: Detrended Regression Plot for 2026-10-31

8 Date: 2026-11-01

Series ID: MRTSMPCSM44X72USN
This series is titled Retail Sales: Retail Trade and Food Services and has a frequency of Monthly. The units are Percent Change from Preceding Period and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1992-02-01 and the observation end date is 2026-08-01.The popularity of this series is 7.

Series ID: ATNHPIUS41660Q
This series is titled All-Transactions House Price Index for San Angelo, TX (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-07-01 and the observation end date is 2026-04-01.The popularity of this series is 1.

8.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ATNHPIUS41660Q	**R-squared:**	0.005
Model:	OLS	**Adj. R-squared:**	-0.002
Method:	Least Squares	**F-statistic:**	0.6811
Date:	Sun, 01 Nov 2026	**Prob (F-statistic):**	0.411
Time:	17:19:26	**Log-Likelihood:**	-804.57
No. Observations:	137	**AIC:**	1613.
Df Residuals:	135	**BIC:**	1619.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	195.0947	8.184	23.839	0.000	178.910	211.280
value_fred_MRTSMPCSM44X72USN	0.6157	0.746	0.825	0.411	-0.860	2.091

Omnibus:	14.744	**Durbin-Watson:**	0.015
Prob(Omnibus):	0.001	**Jarque-Bera (JB):**	17.203
Skew:	0.858	**Prob(JB):**	0.000184
Kurtosis:	2.736	**Cond. No.**	12.2

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

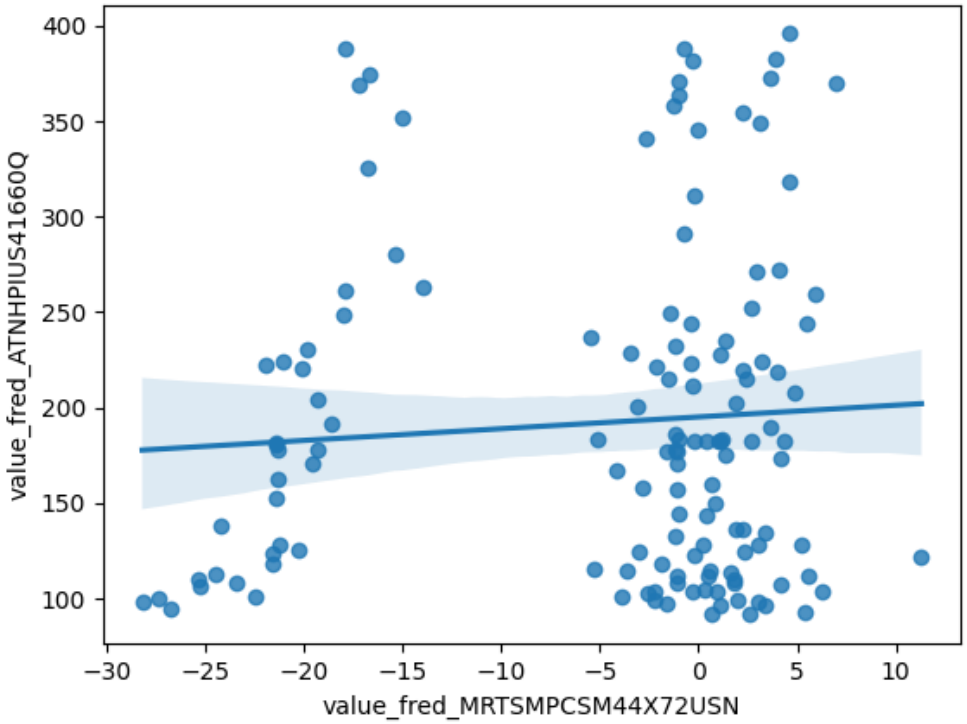


Figure 13: Raw Regression Plot for 2026-11-01

8.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ATNHPIUS41660Q_detrended	**R-squared:**	0.004
Model:	OLS	**Adj. R-squared:**	-0.004
Method:	Least Squares	**F-statistic:**	0.5201
Date:	Sun, 01 Nov 2026	**Prob (F-statistic):**	0.472
Time:	17:19:26	**Log-Likelihood:**	-646.65
No. Observations:	137	**AIC:**	1297.
Df Residuals:	135	**BIC:**	1303.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-3.134e-12	2.336	-1.34e-12	1.000	-4.620	4.620
value_fred_MRTSMPCSM44X72USN_detrended	0.1702	0.236	0.721	0.472	-0.296	0.637

Omnibus:	16.180	**Durbin-Watson:**	0.037
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	16.608
Skew:	0.798	**Prob(JB):**	0.000248
Kurtosis:	2.400	**Cond. No.**	9.90

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

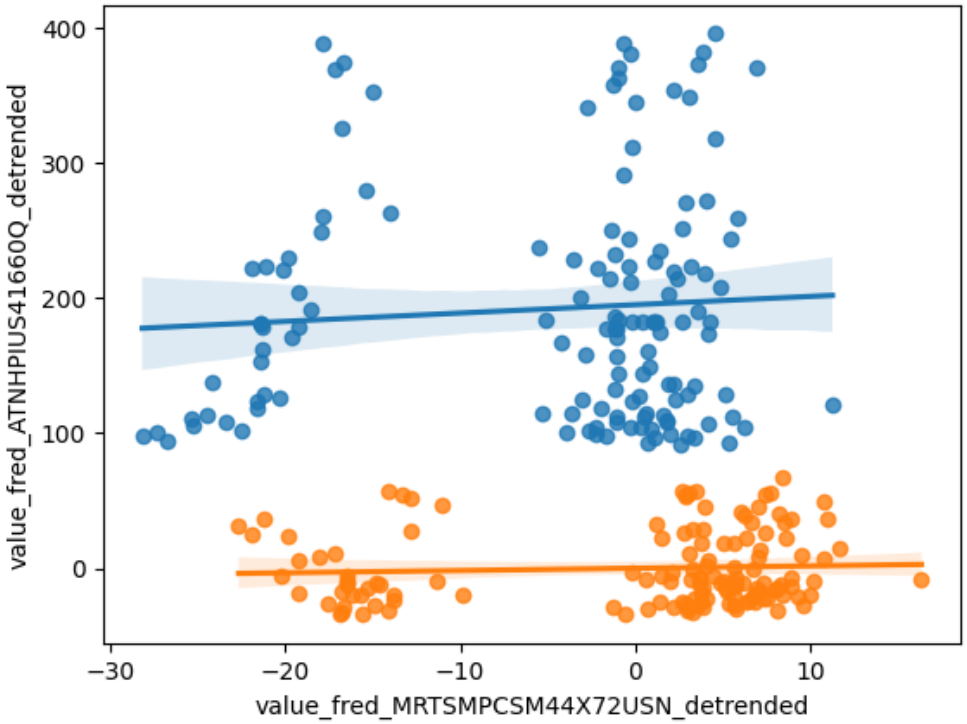


Figure 14: Detrended Regression Plot for 2026-11-01